

PrimeSim™ Continuum Help: What's New

Version W-2024.09, September 2024



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1

What's New in This Release

The *Synopsys Release Notes* provide information on new features, enhancements, changes, known issues, and resolved Synopsys Technical Action Requests (STARs). To view the release notes, go to [My Product Releases](#) on SolvNetPlus, then choose the product and release version of interest.

New Features, Enhancements, and Changes

This release includes the following new features, enhancements, and changes.

- [Analysis Enhancements](#)
- [Device Model Enhancements](#)
- [DP/Cloud Enhancements](#)
- [MOSRA Enhancements](#)
- [Performance Enhancements](#)
- [Signal Integrity Enhancements](#)
- [Enhancement to the HSPICE Device Models](#)
- [Enhancement to the Spectre Simulator Device Models](#)
- [Enhancement to the Eldo Simulator Device Models](#)
- [PrimeSim XA - New, Changed and Obsolete Commands and Options](#)
- [PrimeSim XA - Additional Product and Documentation Enhancements](#)
- [VCS® PrimeSim™ AMS Features and Enhancements](#)
- [VCS PrimeSim AMS - New, Changed and Obsolete Commands and Options](#)
- [VCS PrimeSim AMS - Additional Product and Documentation Enhancements](#)
- [PrimeSim - Enhancement: Additional New or Updated Commands and Options](#)

- [PrimeSim - Additional Product and Documentation Enhancements](#)
- [PrimeSim - GPU Hardware and Software Requirements Updates](#)

Analysis Enhancements

Added the following analysis enhancements:

- Integrated the initialization and main sensitization into a single simulation in PrimeSim HSPICE.
- Added a new option `.option enforce_sensitivity_analysis` to enforce the sensitivity table generation in post process of Monte Carlo analysis (that is, AVA report), if independent random variables (IRV) number is more than 50,000. For more information, see `.OPTION ENFORCE_SENSITIVITY_ANALYSIS` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added the support of expressions and `get_measval()` function in `.DATA` command in PrimeSim HSPICE. For more information, see `.DATA` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Implemented the high/low glitch detection feature based on the vector file in PrimeSim HSPICE.

Device Model Enhancements

Added the following enhancements for device models:

- Added the SiPH related features: optical probing, MTS/3DIC, and so on.
- Enabled thermal modeling in general subckt-based macro model for power devices. For more information, see Simulation of Thermal Effect in Macro Models in the *PrimeSim™ HSPICE® User Guide: Basic Simulation and Analysis*; and `.SET_NODE_TYPE`, `.OPTION ENABLE_SUBCKT_SHE` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added the following new compact models and revisions:
 - BSIM-SOI 100.1. For more information, see LEVEL 70: BSIMSOI Model in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.
 - HiCUM L2 3.1.0. For more information, see LEVEL=8 HICUM Model in the *PrimeSim™ Continuum Reference Manual: Device Models*.
 - HiSIM-SOTB 1.3. For more information, see LEVEL 84: HiSIM-SOTB in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.

- MVSG 4.0. For more information, see LEVEL 82: MVSG Model in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.
- ASM-HEMT 104
- BSIM-CMG106.1+TMEMOD2.1
- BSIM-CMG106.1+TMEMOD2.2
- BSIM-CMG106.1+TMEMOD GIDL equation update
- BSIM-IMG102.96+Aging
- HiSim-HMG update (Beta 5)
- HiSim-HV 2.5.1
- PSP104
- UTSOI 102.8
- Added support for the RRAM compact models in PrimeSim HSPICE.
- Enhanced `MOSRAPRINT` to support `temp` as input.

DP/Cloud Enhancements

Optimized the CDSL application in PrimeSim HSPICE DP for better performance in large Monte Carlo simulations.

MOSRA Enhancements

Added the following MOSRA enhancements:

- Added support to run the MOSRA-based LTE simulation and aging simulation in one SPICE deck in PrimeSim HSPICE.
- Added support for the `RDRMUE` aging simulation in HiSIM-HV 2.1. For more information, see MOSRA API Supported Aging Degradations in the *PrimeSim™ MOSRA User Guide*.
- Enhanced the examples to illustrate the usage of user-defined degradation types. For more information, see Gradual Aging Analysis and MOSRA API Supported Aging Degradations in the *PrimeSim™ MOSRA User Guide*.

Performance Enhancements

Added support for the consistent syntax of bisection with PrimeSim:

- Implemented the same window syntax as in PrimeSim.
- Implemented the one-side search algorithm, to improve the bisection performance.

Signal Integrity Enhancements

Added the following Signal Integrity enhancements:

- Enhanced the edge superposition to capture the jitter amplification accurately in StatEye. For more information, see `.STATEYE` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added the IFFT causality enforcement feature into SPUTIL to enhance the `Causality` function. For more information, see `.CAUSALITY` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added an internal switch to use PrimeSim HSPICE transient engine for StatEye.
- Updated the PrimeSim HSPICE Touchstone demo files, which have significant warnings/errors.

Enhancement to the HSPICE Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new device models:

- MOSFET Models:
 - Level 70, BSIMSOI4 Model, version 100.1
 - Level 73, HiSIM_HV Model, version 2.5.1

For more information, see the Device Models section in *PrimeSim XA User Guide*.

Enhancement to the Spectre Simulator Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new spectre simulator device models:

- MOSFET Device Models: BSIMSOI100 Model

For more information, see the Supported Spectre Simulator Device Models section in *PrimeSim XA User Guide*.

Enhancement to the Eldo Simulator Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new eldo simulator device models:

- Eldo MOSFET Models: Level 72, BSIMSOI4.0 to 4.3 Model
- Eldo MOSFET Models: Level 79, BSIMSOI4.4 to 4.6 Model, BSIMSOI100 Model

For more information, see the Supported Eldo Simulator Device Models section in *PrimeSim XA User Guide*.

PrimeSim XA - New, Changed and Obsolete Commands and Options

The following table lists the commands and command-line options that are new or changed in this release.

Table 1 PrimeSim XA - New and Changed Commands in This Release

Command	Description
<code>.dd_check_block_power</code>	Checks the block power.
<code>iset_speed_scale</code>	Dynamically scales the transient simulation engine tolerance in interactive mode.
<code>cck_analog_pdown</code>	Added new <code>-idsth hizdev1 <value></code> argument that specifies a separate current threshold for paths that contain HiZ-gated MOSFETs.

For more information, see the *PrimeSim™ XA User Guide* and *PrimeSim™ XA Command Reference*.

PrimeSim XA - Additional Product and Documentation Enhancements

- Added the `vcd2vec` command information in the *PrimeSim™ XA Command Reference* manual.

VCS® PrimeSim™ AMS Features and Enhancements

In version W-2024.09, the VCS PrimeSim AMS tool provides the following enhancements:

VCS PrimeSim AMS - New, Changed and Obsolete Commands and Options

The following table lists the commands and command-line options that are new or changed in this release.

Table 2 *New and Changed Commands in This Release*

Command	Description
<code>udn_bidir</code>	Controls the behavior of an <code>udn_bidir</code> interface element. This command sets a Verilog-A module for a specified user-defined nettype on a specific node.
<code>udn_e2n</code>	Added new arguments <code>-min_delta value</code> and <code>-max_delta value</code> .
<code>udn_n2e</code>	Added new argument <code>-min_delta value</code> .

For more information, see the *VCS® PrimeSim™ AMS User Guide*.

VCS PrimeSim AMS - Additional Product and Documentation Enhancements

- Added the following new topics in the *VCS® PrimeSim™ AMS User Guide*:
 - Introducing the VCS PrimeSim AMS Tool
 - Running Cosimulation
 - Pausing and Resuming Simulations
 - Debugging Current Consumption

Following are the new sections that are added to the “User-Defined Nettypes With Custom Interface Elements” topic of *VCS® PrimeSim™ AMS User Guide*.

- Placement of the Verilog-A Interface Elements
- Resolution Function
- Waveform Names

- Example of Unidirectional and Bidirectional Interface Elements
- Commands to Change the Arguments of the Interface Elements

PrimeSim - Enhancement: Additional New or Updated Commands and Options

Additional new or updated PrimeSim netlist options and commands:

- Added `primesim_cck_report_limit` option - Controls the maximum number of error reports generated for each `file.chkdevop` file during simulation.
- Added `primesim_output_op` option - Sets the output format for the operating point (.op) analysis.
- Added `primesim_spf_include` option - Includes only the specified instances from SPF back-annotation and while ignoring the rest of the instances
- Added `primesim_analog` option - Provides a turnkey accuracy setting to run large post-layout analog centric circuits in PrimeSim Pro.
- Added multi-CPU command-line option `-dpincremental` - Reruns a Monte Carlo simulation for unfinished or new samples. After rerun, it automatically combines the measure results from the original runs.
- PrimeSim command line argument `-dscale` is enhanced to support a new usage format as:

```
primesim -dscale output.dynamic -dp number
```

Additional new or updated PrimeSim SPICE netlist commands and control options:

- Updated the `.NOISE` command with new argument `SEPARATE_NOISE` that specifies whether to separate output noise contribution from any noise source into input noise and noise gain.

For more information, see the PrimeSim™ User Guide: Pro and SPICE.

PrimeSim - Additional Product and Documentation Enhancements

- Enhanced PrimeSim to support `PSFXL` output format for transient analysis.
- Added support for `zstd` compression file format.
- PrimeSim SPICE out-of-the-box (OOTB) enhancements on:
 - Transient simulation performance
 - S-parameter simulation robustness, performance, and memory usage

- DCOP robustness, ease of use convergence
- Small-signal analysis robustness and performance
- Updated “Distributed Processing Configuration Files” table in “Running Distributed Processing on a Network Grid” topic to add information related to “CUSTOM” protocol.
- Updated “Ease-of-Use Settings for runlv” and “PrimeSim runlv for analog circuit simulation” tables in “PrimeSim SPICE Accuracy and Speed Options (-runlv and .option runlv)” topic.
- Added `primesim_case_sync_in_subckt` option information in the user guide.
- Added `-spectre` command-line option information in the user guide.

For more information, see the PrimeSim™ User Guide: Pro and SPICE.

PrimeSim - GPU Hardware and Software Requirements Updates

In version W-2024.09, recommended GPU driver and OS have been updated:

- GPU Driver: 535.183.01 and newer versions
- OS: AlmaLinux 8.6 and newer versions

For more information, see the GPU Hardware and Software Requirements in PrimeSim™ User Guide: Pro and SPICE.